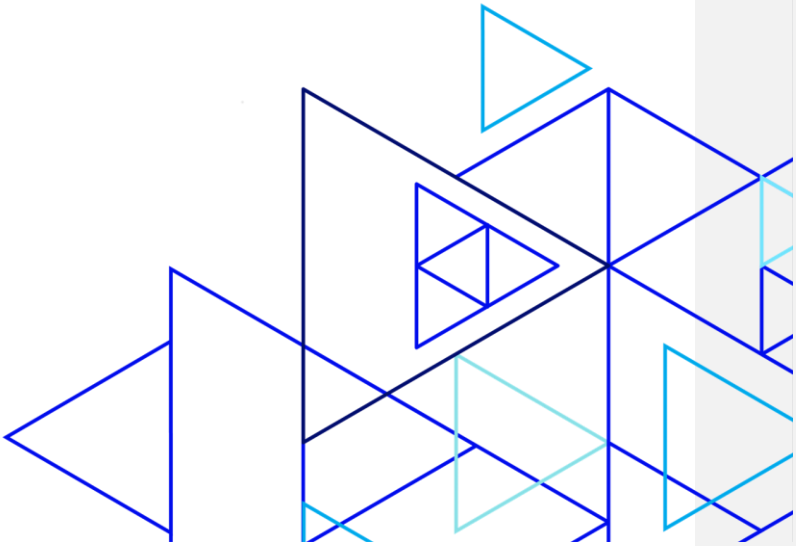
OVHcloud

First Steps on OPCP

Basic networking & compute



Revision History

Version	Author	Date	Notes
0.0	Apolline Fouquet	22/12/2025	Creation
0.1	Apolline Fouquet	29/12/2025	Published draft on team repository
1.0	Apolline Fouquet	08/01/2026	Adding ironic, update nova and precisions on network & SSH
1.1	Samuel LEPETRE	03/03/2026	Adding commands for potential easy copy/paste

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Introduction

The purpose of this document is to give you the keys to start using OPCP for simple use cases.

OpenStack Components

The two main components which are used by OpenStack while you perform your first steps are Neutron and Nova.

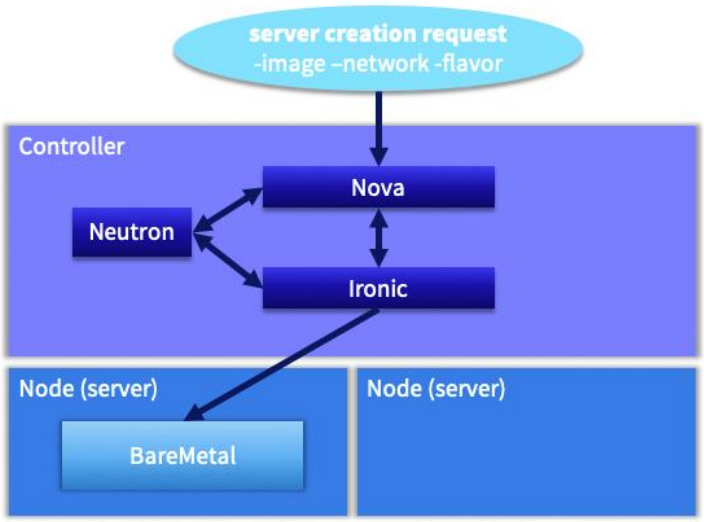
Neutron

Neutron provides virtual networking. Its role is to configure and orchestrate the virtual network resources and interfaces on top of physical networking infrastructure, to enable the requested network.

Nova

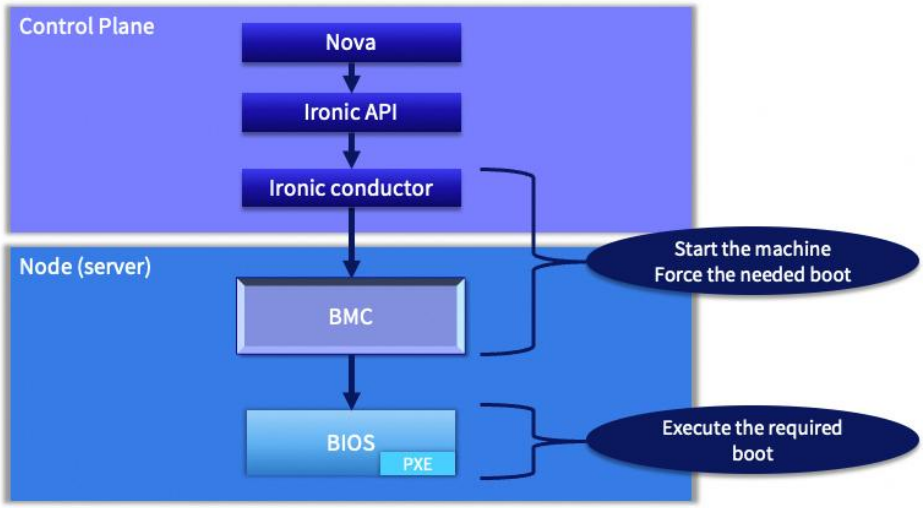
Nova is the orchestrator which enables the creation and management of servers/baremetal. Its role is to:

- get the available nodes,
- calculate which node is the best candidate to host the requested server, based on pre-defined rules and filters and reserve the chosen node
- Tell ironic to prepare a baremetal meeting the requirements



Ironic

Ironic is the component which has the responsibility to provision the server/baremetal accordingly with the requirement's Nova received from the user. Ironic speaks to the BMC (a controller which can control parts of the internal system), can order to power on the physical server and force a specific boot.


Control Plane
Nova
Ironic API
Ironic conductor
Node (server)
BMC
BIOS
PXE
Start the machine
Force the needed boot
Execute the required
boot

Access OPCP

The first step to access OPCP is to create and activate a virtual environment at the location of your choice. This allows to locally set up the environment you will need to use OPCP.

```
afouquet@C02GT0P1Q6LW OPCP % python3 -m venv myvenv
```

```
afouquet@C02GT0P1Q6LW OPCP % source myvenv/bin/activate
```

python3 -m venv myvenv
source myvenv/bin/activate

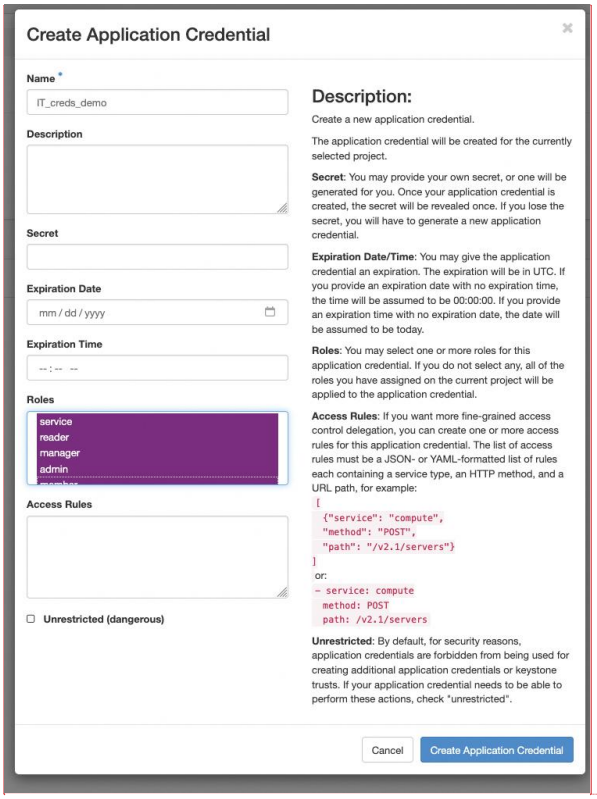
You need to download the *python-openstackclient* library.

```
(myvenv) afouquet@C02GT0P1Q6LW OPCP % pip install -U python-openstackclient
```

MAC : pip install -U python-openstackclient
LINUX: pip install openstackclient

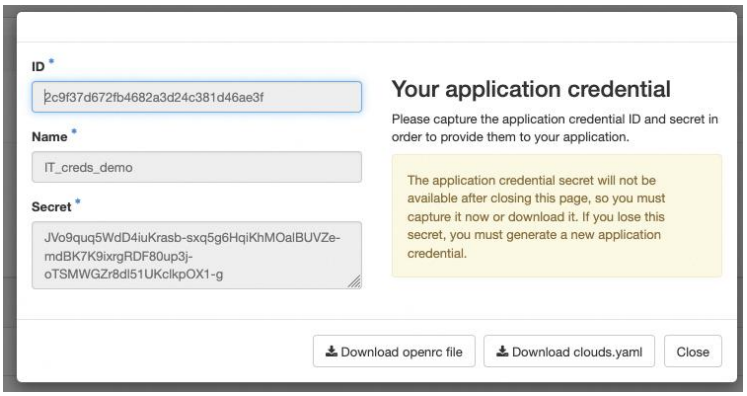
Generate your Application Credentials on the Admin Dashboard, these will allow you to authenticate to OpenStack.

You can select the roles your user will need.



Commenté [AF1]: à update avec méthode de Sam

After the validation of your configuration, you will get this pop-up. You need to download the openrc file.



This file, containing the full environment configuration to access OpenStack, can be copied in the same folder as your *venv*. You need to execute it, to apply the environment variables with the ***source*** command.

Finally, use the ***openstack token issue*** to test your connection and see your token. If you are successfully authenticated, you should get the result as show in figure below.

```
(myvenv) afouquet@C02GT0P1Q6LW OPCP % source /Users/afouquet/Nextcloud/Document/OPCP/app-cred-it_creds-openrc.sh
(myvenv) afouquet@C02GT0P1Q6LW OPCP % openstack token issue
+------------+----------------------
```

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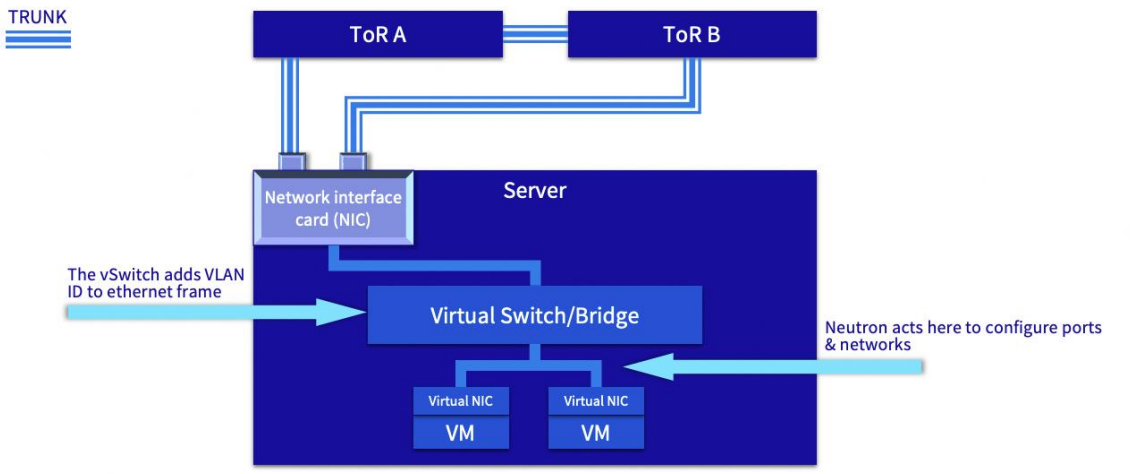
Create network

Before provisioning a server into an OPCP project, you need to create a network first. The diagram below provides an overview of the network conception for OPCP.

Commenté [AF2]: Schéma à remplacer --> validé mais pas bon dans OPCP



The figure below shows the command to use in OPCP to create a network.

```
(myvenv) afouquet@C02GT0P1Q6LW OPCP % openstack network create --project lab-apo --provider-network-type vlan --provider-physical-network physnet1 --provider-segment 2955 my_networklab_a
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2025-12-22T13:43:45Z                 |
| description               |                                      |
| dns_domain                | None                                 |
| id                        | c29c4e15-958e-46ff-919B-ea087a37a967 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | False                                |
| is_vlan_qinq              | None                                 |
| is_vlan_transparent       | None                                 |
| mtu                       | 1500                                 |
| name                      | my_networklab_a                      |
| port_security_enabled     | True                                 |
| project_id                | bbb2fcf02f23415ab8019e4c94c26f81     |
| provider:network_type     | vlan                                 |
| provider:physical_network | physnet1                             |
| provider:segmentation_id  | 2955                                 |
| qos_policy_id             | None                                 |
| revision_number           | 1                                    |
| router:external           | Internal                             |
| segments                  | None                                 |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      |                                      |
| updated_at                | 2025-12-22T13:43:45Z                 |
+---------------------------+--------------------------------------+
```

openstack network create --project lab-pro --provider-network-type vlan --provider-physical-network physnet1 --provider-segment 2955 my_network_lab_a

With OPCP we use VLAN, with trunk connections to enable multi-tenant functionalities. This is the *provider network type* parameter. The VLAN ID is the *provider segment* parameter. The VLAN ID must be more than 200 (the first ones are used by the system). You can use the *openstack network list --long* command to see which IDs are already used.

~~Network segment list mais ne marche pas !~~

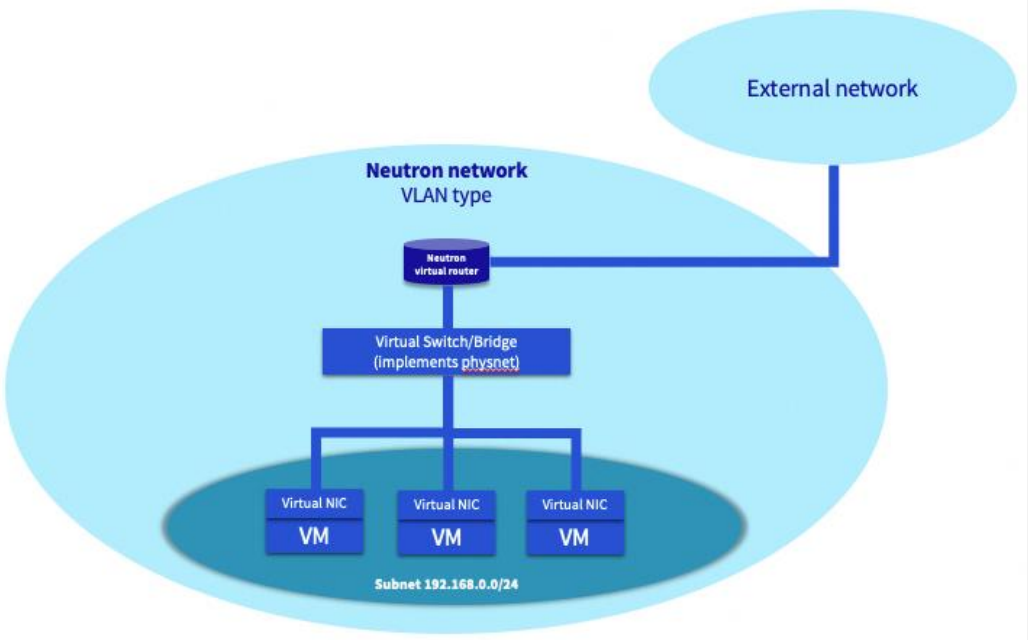
A physical network needs to be specified in *--provider-physical-network*. The default is **physnet1**. This designates the virtual switch which will handle the traffic of the network you are creating.

If you want your network to be shared between multiple projects, you can add the *--share* command.

Create subnet

To enable communication within the network you just created, it is needed to create a subnet to handle IP addresses when there is an external connection. The diagram below is the representation of the network with the enabled level 3 communication.

Commenté [AF3]: Schéma validé mais à changer sur contexte OPCP



You can use the command below.

```
(myvenv) afouquet@C02GT0P1Q6LW OPCP % openstack subnet create --network my_networklab_a --subnet-range 192.168.50.0/24 --allocation-pool start=192.168.50.10,end=192.168.50.70 subnet_lab_a
+-----------------------+--------------------------------------+
| Field                 | Value                                |
+-----------------------+--------------------------------------+
| allocation_pools      | 192.168.50.10-192.168.50.70          |
| cidr                  | 192.168.50.0/24                      |
| created_at            | 2025-12-22T15:39:07Z                 |
| description           |                                      |
| dns_nameservers       |                                      |
| dns_publish_fixed_ip  | None                                 |
| enable_dhcp           | True                                 |
| gateway_ip            | 192.168.50.1                         |
| host_routes           |                                      |
| id                    | 61393b13-f1b1-43d5-a287-b0bd6efd41c1 |
| ip_version            | 4                                    |
| ipv6_address_mode     | None                                 |
| ipv6_ra_mode          | None                                 |
| name                  | subnet_lab_a                         |
| network_id            | c29c4e15-958e-46ff-9198-ea087a37a967 |
| project_id            | 2a50a44794b94c3696f92b47d61f8dc6     |
| revision_number       | 0                                    |
| segment_id            | None                                 |
| service_types         |                                      |
| subnetpool_id         | None                                 |
| tags                  |                                      |
| updated_at            | 2025-12-22T15:39:07Z                 |
+-----------------------+--------------------------------------+
```

openstack subnet create --network my_network_lab_a --subnet-range 172.68.50.0/24 --allocation-pool start= 172.68.50.10,end= 172.68.50.70 subnet_lab_a

You first need to specify the network you just created with the *--network* parameter. Then, you must choose the IP range of your subnet with *--subnet-range.*

Finally, you have the **option** to specify an *--allocation-pool* with *start* and *end*. It allows to specify which range of IP can be used for attribution to the server instances. The other IP addresses are reserved for other purposes to suit your needs.

As a reminder, private IP addresses are local and can't be routed on the internet. They are defined in the reserved ranges 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. A public IP address can be reached from outside of the local network.

The subnet mask (ex: '/24') determines the number of IP addresses allocated for the machines using your network. The larger the mask, the smaller the network is. For example : /24 allocates 256 addresses, with /26 it's 64.

With these two simple steps, you have now enabled layer 2 & 3 communications for the servers you will be instantiating.

~~Configure quota~~

~~Show quota of a project~~

```
(myvenv) afouquet@C02GT0P1Q6LW OPCP % openstack quota show lab-apo
+----------------------------+---------+
| Resource                   |   Limit |
+----------------------------+---------+
| cores                      |     192 |
| instances                  |      10 |
| ram                        | 1536000 |
| fixed_ips                  |    None |
| networks                   |     100 |
| volumes                    |      10 |
| snapshots                  |      10 |
| gigabytes                  |    1000 |
| backups                    |      10 |
| volumes_public_standard    |      -1 |
| gigabytes_public_standard  |      -1 |
| snapshots_public_standard  |      -1 |
| groups                     |      10 |
| trunk                      |      -1 |
| check_limit                |    None |
| health_monitors            |    None |
| listeners                  |    None |
| load_balancers             |    None |
| l7_policies                |    None |
| pools                      |    None |
| ports                      |     500 |
| project_id                 |    None |
| rbac_policies              |      10 |
| routers                    |      10 |
| subnets                    |     100 |
| subnet_pools               |      -1 |
| injected-file-size         |   10240 |
| injected-path-size         |     255 |
| injected-files             |       5 |
| key-pairs                  |     100 |
| properties                 |     128 |
| server-group-members       |      10 |
| server-groups              |      10 |
| floating-ips               |      50 |
| secgroup-rules             |     100 |
| secgroups                  |      10 |
| backup-gigabytes           |    1000 |
| per-volume-gigabytes       |      -1 |
+----------------------------+---------+
```

~~Documentation de chaque field ? Très précis ce résultat~~

Authenticate your machine

To use the server you will be creating, you need to have an SSH key pair. If you already have one, you can skip the first step.

Else, you need to go into your root directory to create the .ssh file, using the 2 commands below. The 700 parameter is to enable all the rights on the file you are creating.

```
(myvenv) afouquet@C02GT0P1Q6LW / % mkdir ~/.ssh
(myvenv) afouquet@C02GT0P1Q6LW / % chmod 700 ~/.ssh
```

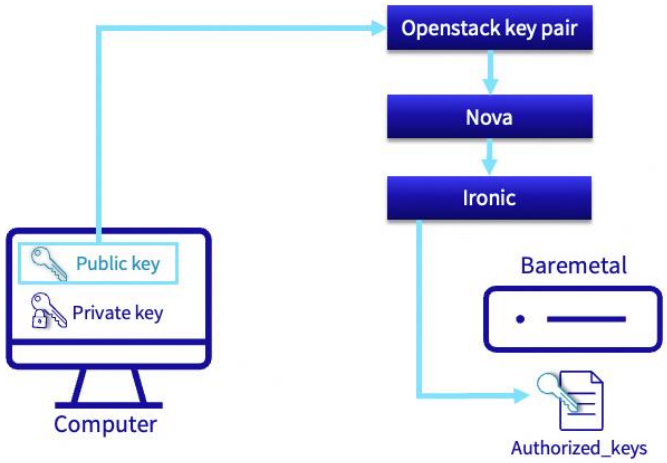
mkdir ~/.ssh
chmod 700

Then, you need to share your public key with OpenStack, allowing it to recognize you so that you can authenticate to your virtual machines.

```
(myvenv) afouquet@C02GT0P1Q6LW OPCP % openstack keypair create --public-key ~/.ssh/id_rsa.pub lab-key
+-------------+-------------------------------------------------------------------+
| Field       | Value                                                             |
+-------------+-------------------------------------------------------------------+
| created_at  | None                                                              |
| fingerprint | 94:82:87:52:f1:b4:0a:29:63:a5:a3:3c:e3:ed:e3:17                   |
| id          | lab-key                                                           |
| is_deleted  | None                                                              |
| name        | lab-key                                                           |
| type        | ssh                                                               |
| user_id     | 5c656f47f6f6191d0126f40de0b4a5bf90f80de5a4e47926053bc7d1fac3ebbb  |
+-------------+-------------------------------------------------------------------+
(myvenv) afouquet@C02GT0P1Q6LW OPCP % openstack keypair list
+--------+-------------------------------------------------+------+
| Name   | Fingerprint                                     | Type |
+--------+-------------------------------------------------+------+
| lab-key | 94:82:87:52:f1:b4:0a:29:63:a5:a3:3c:e3:ed:e3:17 | ssh  |
+--------+-------------------------------------------------+------+
```

openstack keypair create --public-key ~/.ssh/id_rsa.pub test_enduser_key

The diagram below explains why you need to share your public key to openstack and its path to provide you access to the server you will be requesting.



Creating an instance

Now that your environment is ready, you can create an instance. These steps can be done by end-users.

When creating a server, you can specify a flavor and an image. The flavor is a hardware profile provided as a template for the machine, and the images are the OS and software templates.

You can list both with the following commands, to get the available templates.

```
openstack flavor list
+--------------------------------------+---------------------+---------+------+-----------+-------+-----------+
| ID                                   | Name                |     RAM | Disk | Ephemeral | VCPUs | Is Public |
+--------------------------------------+---------------------+---------+------+-----------+-------+-----------+
| 4e731d33-4c1e-4286-bc63-68c2cba3bc59 | gpu                 |     400 |  900 |         0 |   128 | True      |
| 8bc6b1e0-dfdd-44d8-83fb-cef5e8fb6ec2 | scale-2             |     256 | 4000 |         0 |    32 | True      |
| f975e49a-fd75-4a7e-868e-2842972e82e2 | hsm                 |     512 | 4000 |         0 |   128 | True      |
| fcd2c556-9a16-4532-a5c8-e9d9275ba78f | scale-1             |     128 | 2000 |         0 |    24 | True      |
| ff5b4d7d-46c9-4bf5-b300-f1f465661e64 | scale-3             |     256 | 4000 |         0 |    48 | True      |
+--------------------------------------+---------------------+---------+------+-----------+-------+-----------+
```

openstack flavor list

```
openstack image list
+--------------------------------------+--------------------------------------------+--------+
| ID                                   | Name                                       | Status |
+--------------------------------------+--------------------------------------------+--------+
| 6540686f-0150-496b-a894-03d95ef8bc7e | ironic-agent.initramfs                     | active |
| 5f6d4237-023b-4a25-8ceb-7051ffc6d632 | ironic-agent.kernel                        | active |
| 7600f9fc-fc1f-4fb8-aaa3-878bb9399d26 | CentOS 9 OPCP                              | active |
| ebabdaaa-a52d-4246-9790-e77ba5c49519 | Debian 12 LVM OPCP                         | active |
+--------------------------------------+--------------------------------------------+--------+
```

openstack image list

Choose an image and flavor which best suit your needs and inject them in the command below, with the key you just shared with OpenStack and the network you created earlier. At the end of the command, you should specify a name for your instance.

```
(myvenv) afouquet@C02GT0P1Q6LW OPCP % openstack server create --key-name lab-key --flavor gpu --image "Debian 13 BMPOD" --network my_networklab_a lab_firstserver
+-------------------------------------+-----------------------------------------------
```

You can list the servers to check the status of the one you created.

```
openstack server list
+--------------------------------------+-----------------+--------+-------------------------+---------------------+---------+
| ID                                   | Name            | Status | Networks                | Image               | Flavor  |
+--------------------------------------+-----------------+--------+-------------------------+---------------------+---------+
| 2f9c2e41-b4dc-4787-a3ef-9b2b2d686dbb | OPCPdocs-server | ACTIVE | opcpdocs=192.168.120.59 | Debian 12 LVM OPCP  | scale-1 |
+--------------------------------------+-----------------+--------+-------------------------+---------------------+---------+
```

openstack server list

Troubleshooting

If you encounter an issue with the status of the server you tried to provision, you can use the first command below with your server ID to get the ID of your creation request. Then use the second command to get the details on your request.

```
(myvenv) afouquet@C02GT0P1Q6LW OPCP % openstack server event list e1e91de1-a20e-4d79-828d-1b5c2c53fc08
+------------------------------------------+--------------------------------------+--------+----------------------------+
| Request ID                               | Server ID                            | Action | Start Time                 |
+------------------------------------------+--------------------------------------+--------+----------------------------+
| req-e651a79f-dbae-46aa-96fb-888d2de86bea | e1e91de1-a20e-4d79-828d-1b5c2c53fc08 | create | 2025-12-24T13:58:06.022940 |
+------------------------------------------+--------------------------------------+--------+----------------------------+
(myvenv) afouquet@C02GT0P1Q6LW OPCP % openstack server event show e1e91de1-a20e-4d79-828d-1b5c2c53fc08 req-e651a79f-dbae-46aa-96fb-888d2de86bea
+-------------+-------------------------------------------
```